3. In Fig. 3, the left MIM capacitance is C_S; the right MIM has capacitance C_D. In Fig. a, Q_1 , Q_2 are charges on C_S, C_D. Thus the net island charge is given by

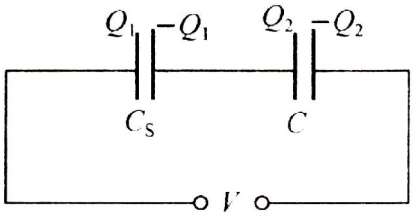


Fig. a

$$-ne = Q_2 - Q_1 . \qquad (11)$$

The electrostatic energy is the sum of that in C_S and C_D:

$$U = \frac{Q_1^2}{2C_S} + \frac{Q_2^2}{2C_D} ; \qquad (12)$$

The voltage is

$$V = \frac{Q_1}{C_S} + \frac{Q_2}{C_D} . \qquad (13)$$

Solving (11) ~ (13) gives

$$U = \frac{1}{2}CV^2 + \frac{(Q_2 - Q_1)^2}{2(C_S + C_D)} . \qquad (14)$$

With constant V, (13) gives the energy due only to net island charge: $U_n = (-ne)^2 / 2(C_S + C_D)$.

4. The plot of U_n against $C_G V_G$ is Fig. b; Table 2 gives the $C_G V_G / e$ ranges.

Table 2

n	0	1	2	3
$C_G V_G / e$ range	0～0.5	0.5～1.5	1.5～2.5	2.5～3.0

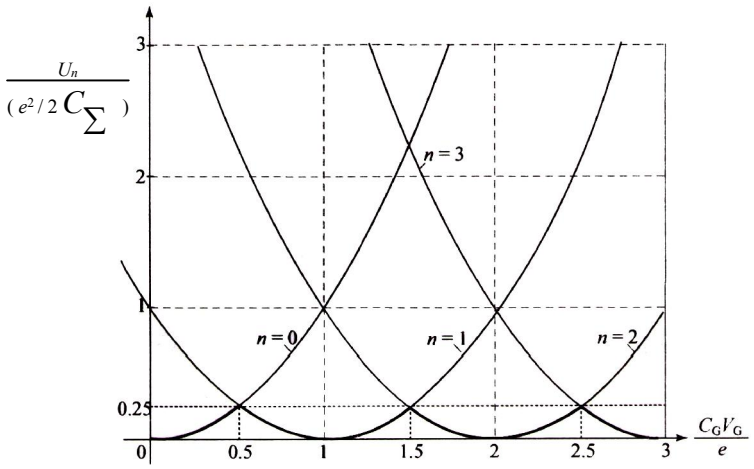


Fig. b